Data Mining

Clustering: Part 4

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Outline

- **Overview of Clustering**
- **Major Clustering Approaches**
  - **K-means Clustering**
  - **Hierarchical Clustering**
  - **DBSCAN Clustering**
- **Cluster Evaluation**

Outline

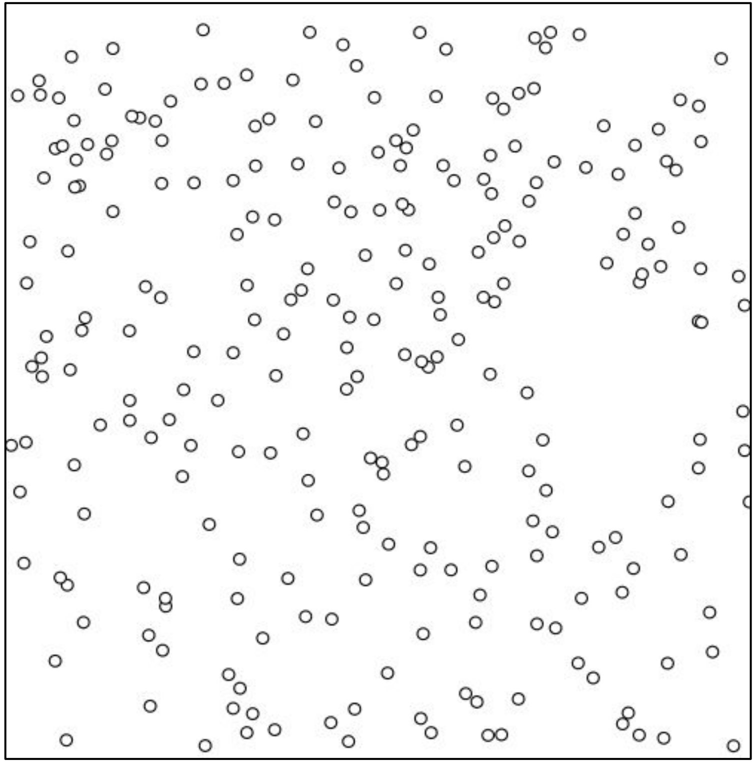
- **Cluster Evaluation**
  - Why Cluster Evaluation?
  - Types of Cluster Evaluation
- Unsupervised Evaluation
  - Cohesion vs Separation
  - Silhouette Coefficient
- Supervised Evaluation
  - Entropy
  - Precision, Recall, F-measure

Why Cluster Evaluation?

- Generate a random data points.
- Data without any structure

Question:

What if we apply K-Means with K=3?



The following link can be used: [K-Means Animation]

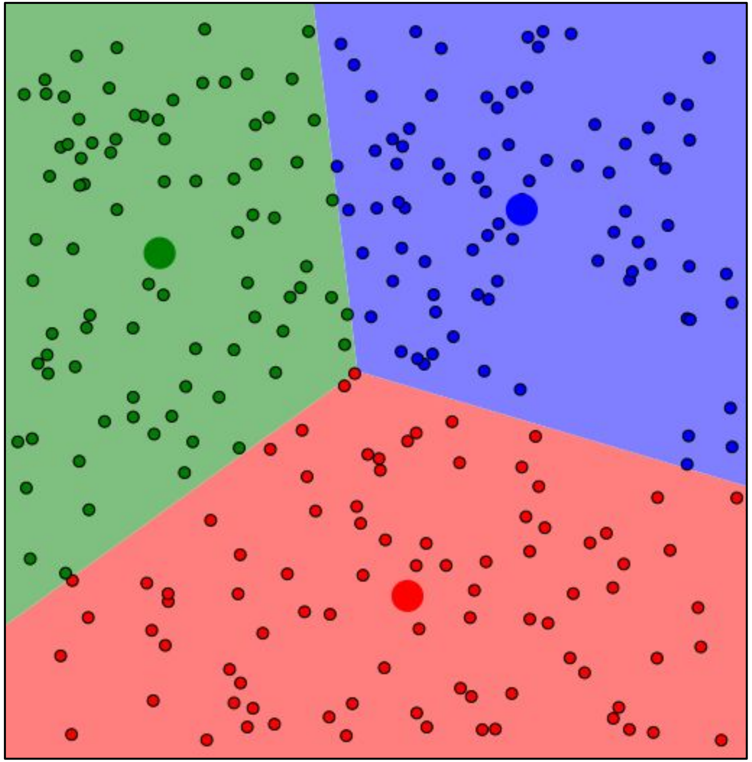
Why Cluster Evaluation?

- Generate a random data points.
- Data without any structure

Results:

Clusters found in random data!

The following link can be used: [K-Means Animation]()



Why Cluster Evaluation?

- **To avoid detecting clusters in random structure**
    - Uncovering whether non-random structure exists in the data.
- **To evaluate clustering results**
    - Assessing how well the clustering aligns with the data without external reference.
- **To compare with external known patterns**
    - Comparing clustering results to externally known information, e.g., class labels.
- **To compare different clusterings and algorithms**
    - Evaluating and comparing different sets of clusters for quality.

“*The validation of clustering structures is the*
most difficult and frustrating part of cluster analysis.

Without a strong effort in this direction,
cluster analysis will remain a black art accessible only to
those true believers who have experience and great courage.”

Algorithms for Clustering Data, Jain and Dubes

Types of Cluster Evaluation

- **Unsupervised (Internal)**: measure the goodness of a clustering structure without respect to external information.
  - The ground truth is not available.
  - **Examples:** Cohesion, Separation, SSE, Silhouette Coefficient.

- **Supervised (External):** measure the extent to which cluster labels match externally supplied class labels.
  - The ground truth is available.
  - **Examples:** Entropy, Precision, Recall, F-measure.

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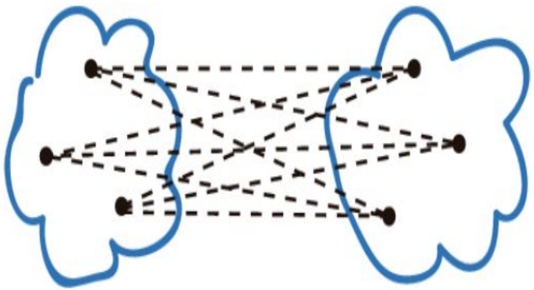
- ❑ Cluster Evaluation
  - ❑ Why Cluster Evaluation?
  - ❑ Types of Cluster Evaluation
- ❑ **Unsupervised Evaluation**
  - ❑ Cohesion vs Separation
  - ❑ Silhouette Coefficient
- ❑ Supervised Evaluation
  - ❑ Entropy
  - ❑ Precision, Recall, F-measure

Cohesion vs Separation

Cluster cohesion (Compactness)

- Measure how closely related object in a cluster.

Cluster Separation



- Measure how distinct or well-separated a cluster is from other clusters.

Cohesion and Separation have two perspectives:

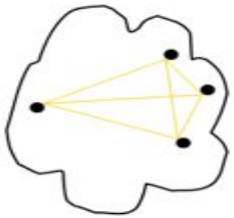
- Graph-based view
- Prototype-based view

Graph-Based View

Weighted graph where the weights are the distances between data points.

- **Cohesion:** Sum of proximities in a cluster.

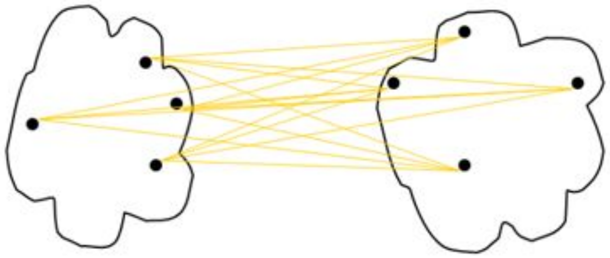
$$cohesion(C_i) = \sum_{\substack{\mathbf{x} \in C_i \\ \mathbf{y} \in C_i}} proximity(\mathbf{x}, \mathbf{y})$$



cohesion

- **Separation:** Sum of proximities between two clusters.

$$separation(C_i, C_j) = \sum_{\substack{\mathbf{x} \in C_i \\ \mathbf{y} \in C_j}} proximity(\mathbf{x}, \mathbf{y})$$



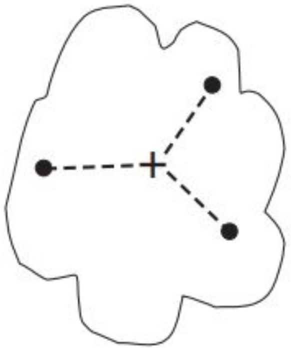
separation

Prototype-Based View

Represent a clusters using their centroids.

- **Cohesion:** Sum of proximities to the cluster centroid.

$$cohesion(C_i) = \sum_{\mathbf{x} \in C_i} proximity(\mathbf{x}, \mathbf{c}_i)$$

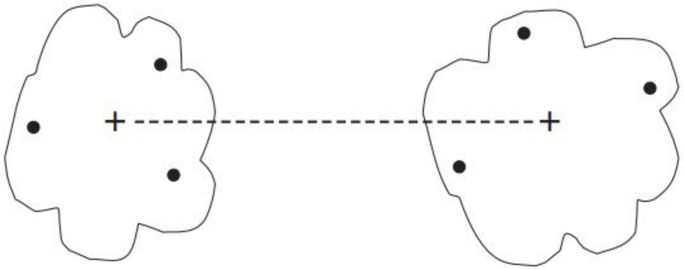


- **Separation:** Sum of proximites between centroids.
  - Between two centroids

$$separation(C_i, C_j) = proximity(\mathbf{c}_i, \mathbf{c}_j)$$

  - Between a cluster centroid and the global centroid.

$$separation(C_i) = proximity(\mathbf{c}_i, \mathbf{c})$$



Prototype-Based View

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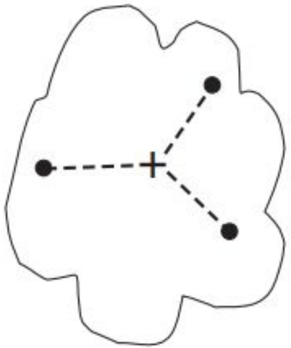
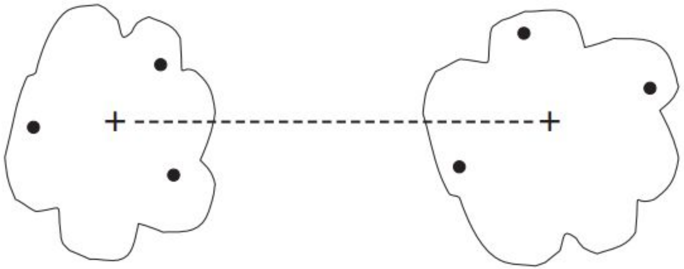


SSE is the sum of prototype-based cohesion of all clusters.



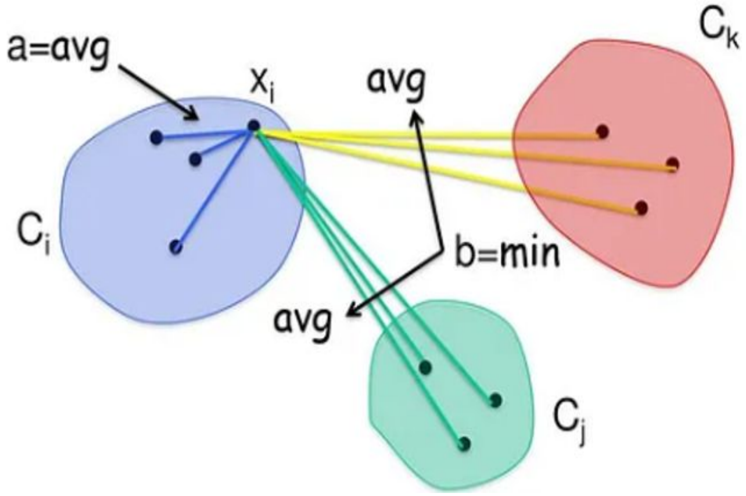
$$separation(C_i) = proximity(\mathbf{c}_i, \mathbf{c})$$

Silhouette Coefficient

- Silhouette coefficient combines **cohesion** and **separation.**
- For an individual point ***i***
  - **a** = average distance of ***i*** to the points in its cluster
  - **b** = ***min***(average distance of ***i*** to points in another cluster)
- The silhouette coefficient for a point is:

  s = (b – a) / max(a, b)

- Value can vary between -1 and 1.
- The closer to 1 the better.
- The silhouette coefficient for a clustering algorithm is the average silhouette coefficient of all individual data points in the dataset.



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  - ❑ Types of Cluster Evaluation
- ❑ Unsupervised Evaluation
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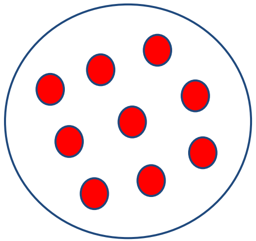
- ❑ **Supervised Evaluation**
  - ❑ Entropy
  - ❑ Precision, Recall, F-measure

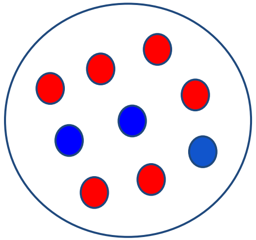
Entropy

Entropy measures the extent to which the clustering structure matches external class labels.



Pure cluster

- Pure cluster is cluster that contain only one class label.
- We measure the purity of a cluster using the entropy.
- How to use entropy for evaluation:
    - Calculate entropy for each cluster.
    - Sum the entropies to get an overall measure.
    - Lower values indicate better alignment with external class labels.



Impure cluster

$$s_c = \sum_{1}^{K} \frac{n_k}{N} s_{Lk} : s_{lk} = \sum_{1}^{L} -p_{Lk} \log_2 p_{lk}$$

N: total number of data points
K: n° of clusters, **L:** n° of class labels
$\mathbf{n_k}$: n° of of data points in cluster k
$\mathbf{p_{lk}}$: proportion of cluster **k**'s points that belong to class **l**.

Entropy: Example

k	p_{1k}	p_{2k}	p_{3k}	s_{Lk}
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0

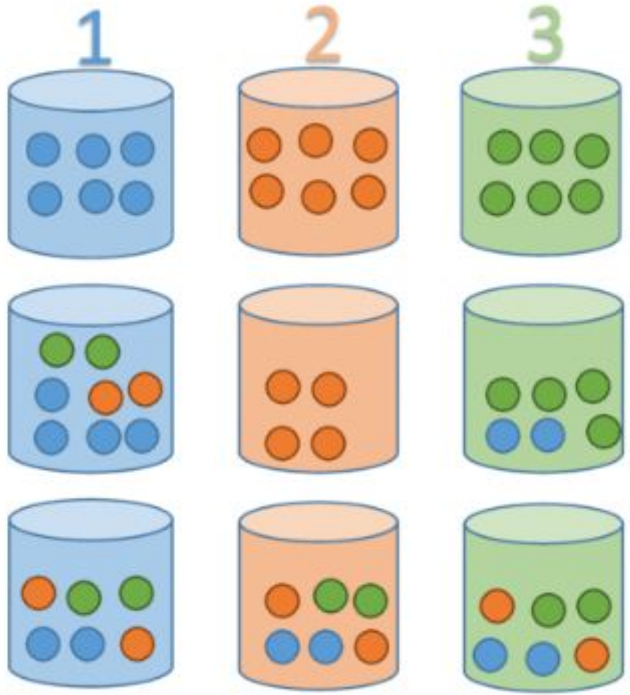
$S_c = 0$

k	p_{1k}	p_{2k}	p_{3k}	s_{Lk}
1	4/8	2/8	2/8	1.5
2	0	1	0	0
3	2/6	0	4/6	0.918

$S_c = 0.971$

k	p_{1k}	p_{2k}	p_{3k}	s_{Lk}
1	2/6	2/6	2/6	1.585
2	2/6	2/6	2/6	1.585
3	2/6	2/6	2/6	1.585

$S_c = 1.585$



$$S_c = \sum_1^K \frac{n_k}{N} s_{Lk} \quad : \quad s_{lk} = \sum_1^L -p_{Lk} \log_2 p_{lk}$$

Precision, Recall, F-measure

- **Precision:** The fraction of a cluster ***i*** that consists of objects of a specified class.

$$\text{Precision}(i,j) = \frac{\text{Number of examples of class } j \text{ in cluster } i}{\text{Size of cluster } i}$$

- **Recall:** The extent to which a cluster contains all objects of a specified class.

$$\text{Recall}(i,j) = \frac{\text{Number of examples of class } j \text{ in cluster } i}{\text{Number of examples of class } j}$$

- **F-measure:** A combination of precision and recall that measures the extent to which a cluster contains only objects of a particular class and all objects of that class.

$$F(i,j) = \frac{2 \times \text{Precision}(i,j) \times \text{Recall}(i,j)}{\text{Precision}(i,j) + \text{Recall}(i,j)}$$